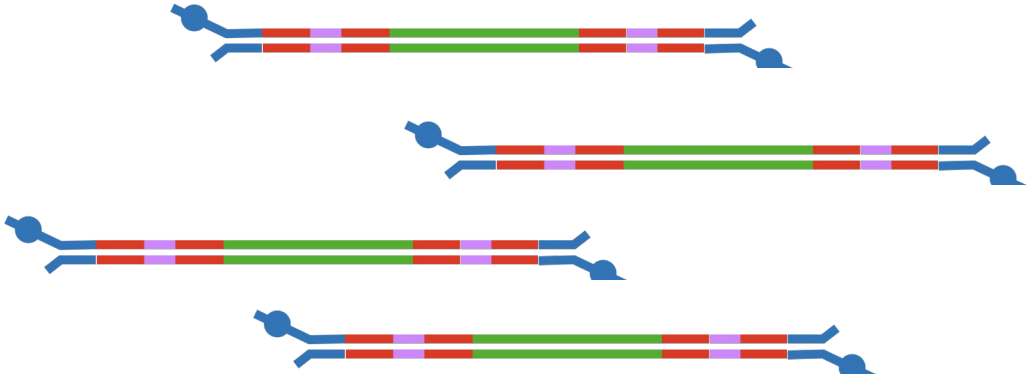
Nanopore RNA and DNA library prep for mNGS

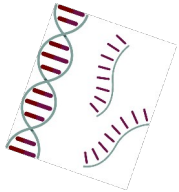


mNGS pipeline

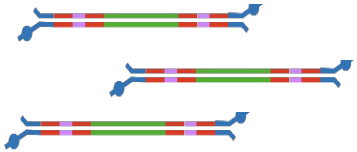
Study design



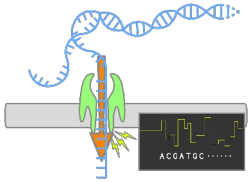
Nucleic acid extraction



Library prep



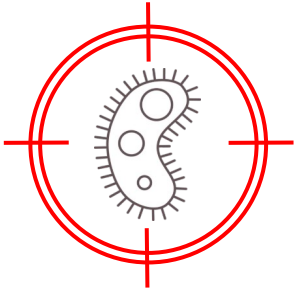
Sequencing



Data analysis



Pathogen ID



Considerations when using Nanopore (ONT) sequencing

Sample integrity

- Important to ensure high quality nucleic acid with little to no degradation
- Optimize sample handling and storage

Library preparation kit (link to available kits)

- DNA or RNA?
- PCR or PCR-free?
- Multiplexing or single sample?

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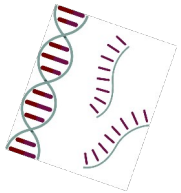
Kit used for ONT RNA mNGS SMART-9N:
Rapid PCR barcoding kit (SQK-RPB114.24)

ONT mNGS pipeline

Study design



xNA extraction



Library prep



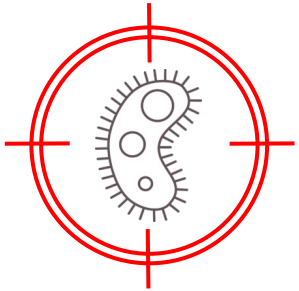
Sequencing



Data analysis



Pathogen ID



Library prep: RNA to sequencing library

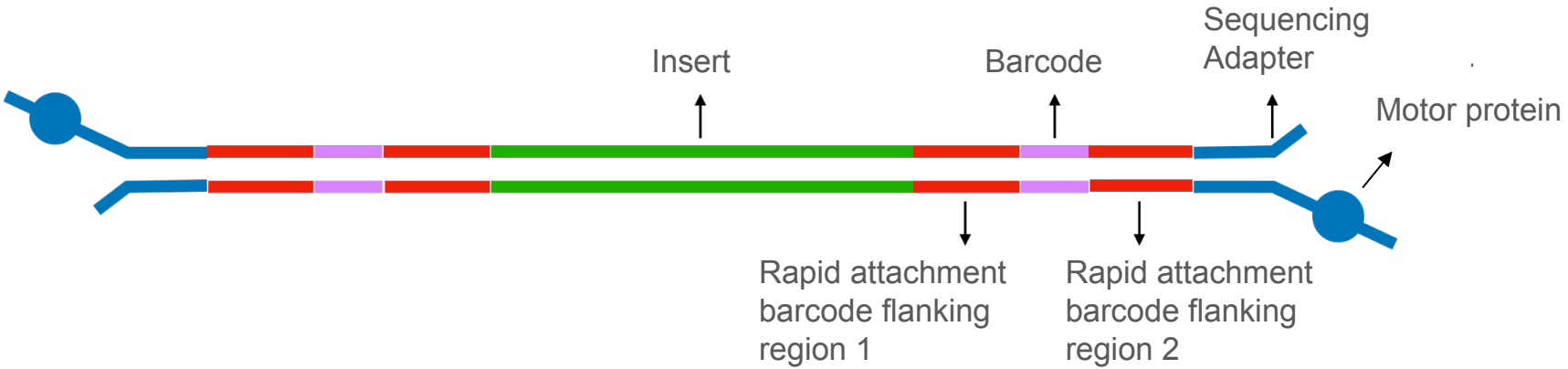
- Sequencing library = Pool of DNA inserts with sequencing adapters attached



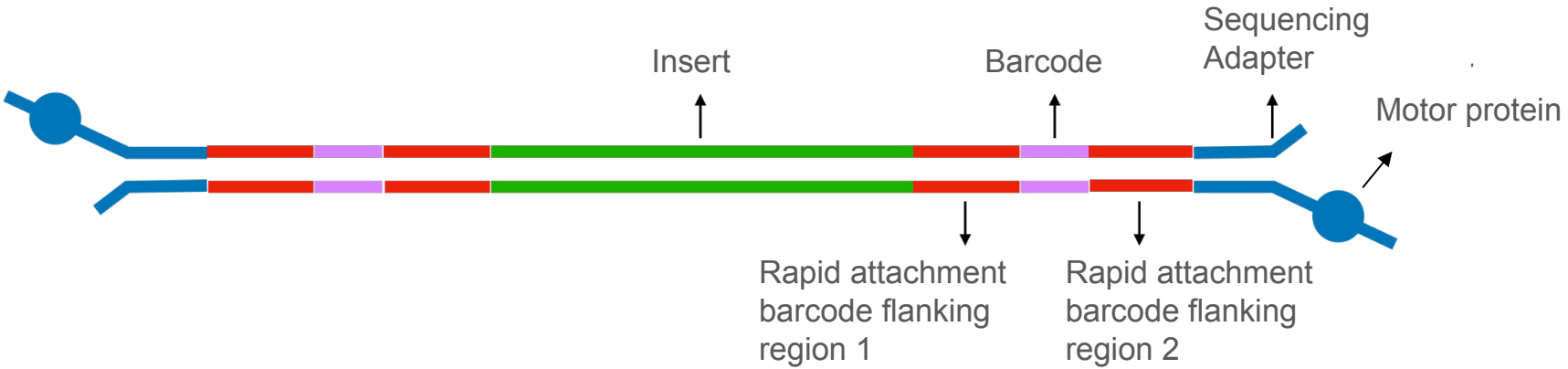
Extracted RNA

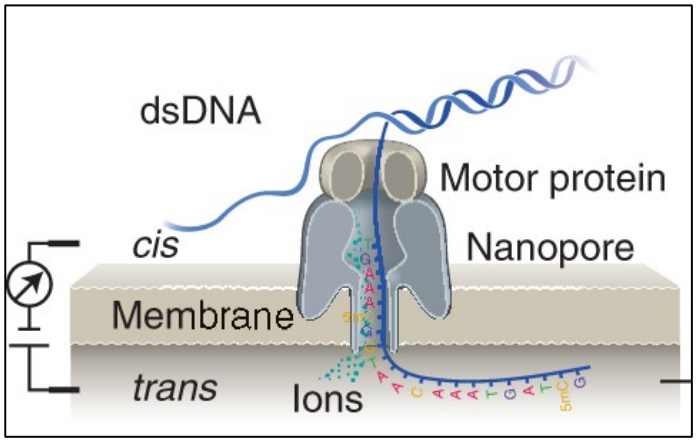
Sequencing library

Anatomy of a completed ONT library

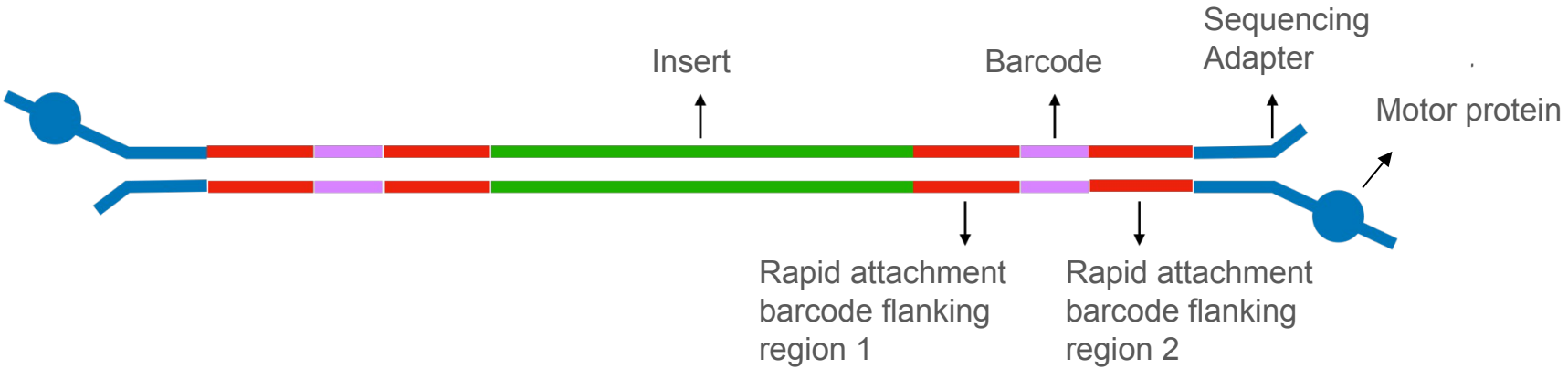


Anatomy of a completed ONT library





Anatomy of a completed ONT library

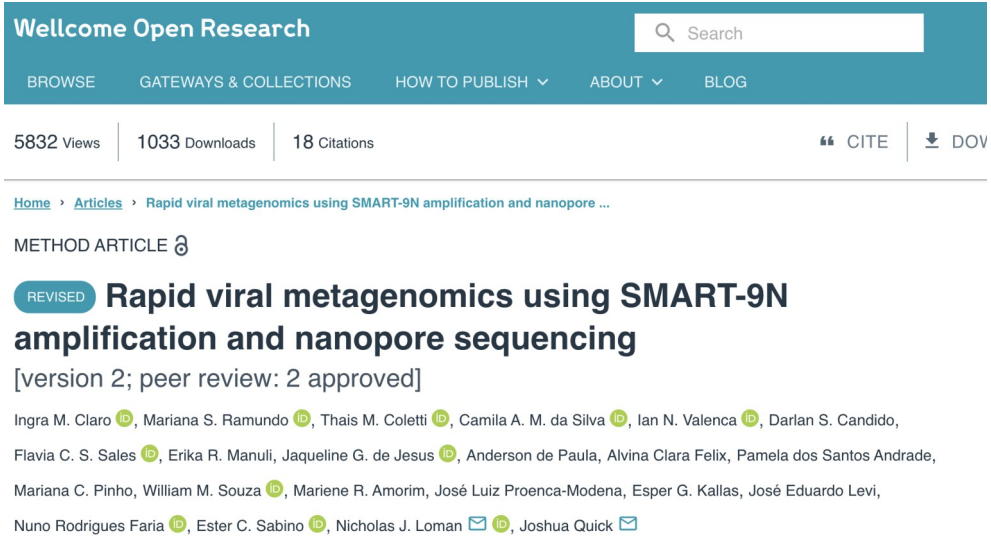


1. **Insert**: cDNA target that will be sequenced
2. **Rapid attachment barcode flanking region 1**: adapter region allows barcode primer to bind for PCR priming; added during cDNA synthesis
3. **Barcode**: DNA barcode specific to each sample; added during PCR amplification
4. **Rapid attachment barcode flanking region 2**: includes special molecule for sequencing adapter ligation; included in barcode primer, added during PCR amplification
5. **Sequencing adapter**: allows binding and movement of DNA through flow cell sequencing pores; includes motor protein; added during sequencing adapter ligation

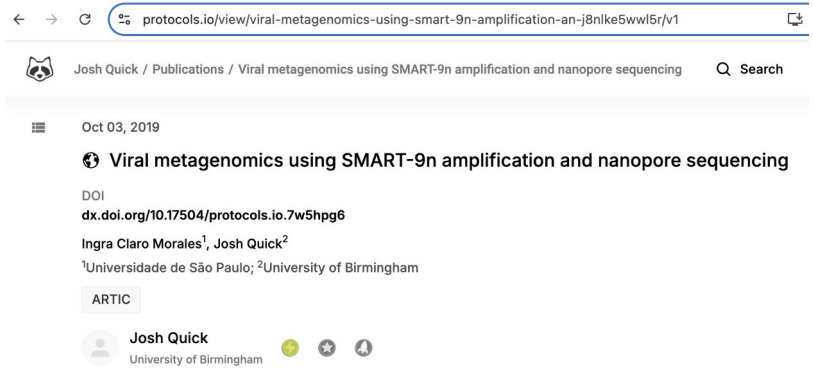
RNA mNGS using Nanopore

For RNA mNGS using Nanopore, we use the SMART-9N protocol with minor modifications

- Increase concentration of oligos for reverse transcription steps
- Adjust annealing temperature for PCR



https://doi.org/10.12688/wellcomeopenres.17170.2

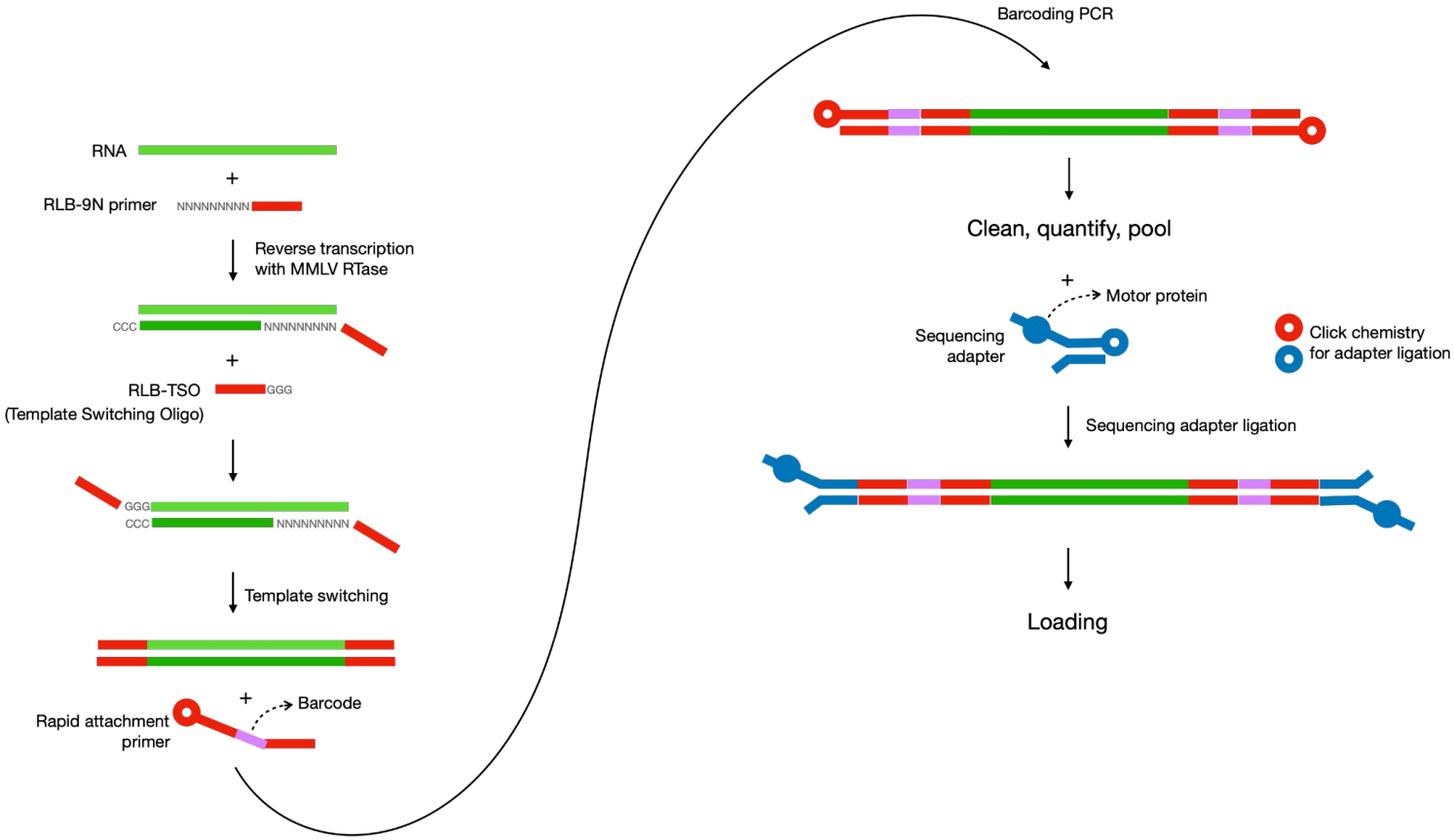


dx.doi.org/10.17504/protocols.io.7w5hpg6

Steps in RNA mNGS library prep for ONT (SMART-9N)

- Annealing
- Reverse transcription + template switching
- Barcoding PCR
- Clean-up, quantification, and pooling
- Sequencing adapter ligation

RNA mNGS library prep for ONT (SMART-9N)



RNA mNGS library prep for ONT (SMART-9N)

STEP 0. Starting material: RNA or TNA

(pre-PCR lab)

- RNA only: move on to annealing and reverse transcription

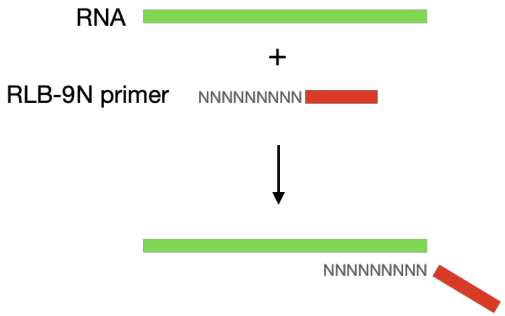
- TNA (total nucleic acid): need to remove DNA
  1. Aliquot sample
  2. DNAse treatment
  3. SPRI bead cleanup (2x)
  4. Qubit quantification

RNA mNGS library prep for ONT (SMART-9N)

STEP 1. Annealing

(pre-PCR lab)

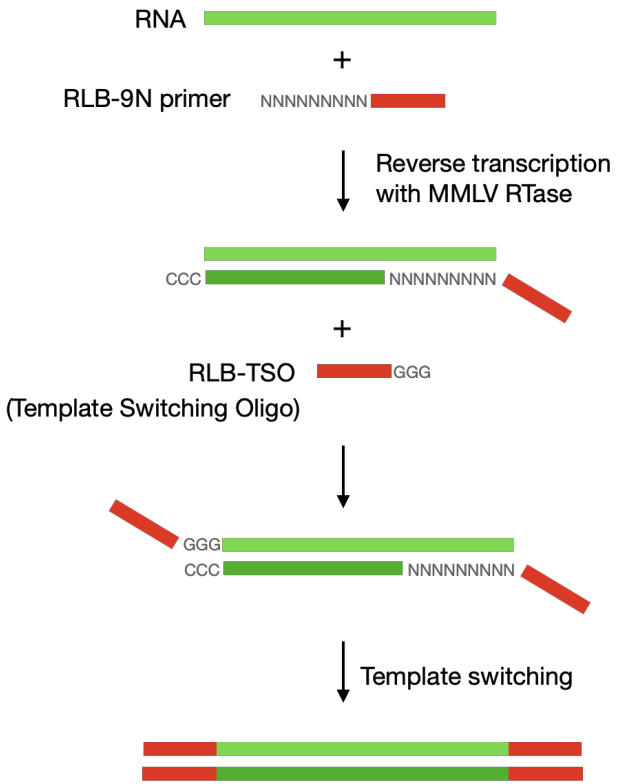
- RLB-9N primer: ONT adaptor with random nonamer
- ONT adaptor sequence corresponds to the transposase adapter used in Rapid PCR Barcoding kit



RNA mNGS library prep for ONT (SMART-9N)

STEP 2. Reverse transcription + template switching (pre-PCR lab)

- RNase OUT + DTT → degrade any RNases present
- Special RTase is used that adds CCC at the end (MMLV)
- RLB-TSO: Template Switching Oligo → anneals to CCC and generates the template switched oligo

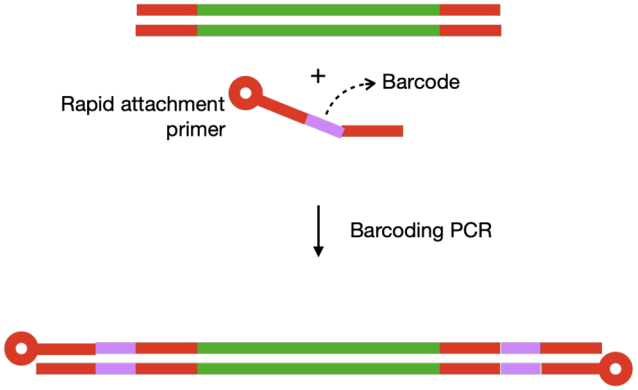


RNA mNGS library prep for ONT (SMART-9N)

STEP 3. Barcoding PCR

(prep pre-PCR lab, amplify in post-PCR)

- Barcoding primers: single barcode, same primer amplifies both ends of each sample
- Barcoding primers have a modified end to facilitate sequencing adapter ligation
- ONT calls these “single rapid attachment primers” and they come with the ONT kit
- PCR conditions:
  - Number of PCR cycles depends on the input; for low input, perform 25 cycles
  - Long extension time used since samples are unfragmented

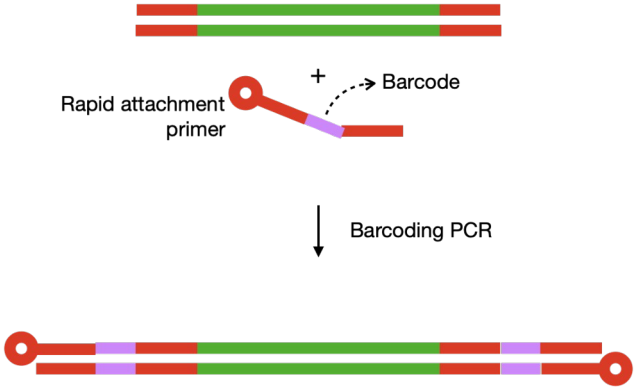


RNA mNGS library prep for ONT (SMART-9N)

STEP 4. Clean-up, QC, and pooling

(post-PCR lab)

- Perform 0.6x SPRI clean-up of libraries after barcoding PCR
- Do QC (quantity and quality)
    - If running Tapestation, use the genomic screentape
    - Ideal size range should be ~1000-7000 bp
- Pooling
    - The goal is to have 100-150 fmol of library in 11µL
    - (This corresponds to a concentration of 10-15 nM)
    - Pooling can be done in nuclease-free water or EB as the kit says (anecdotally, water works better)

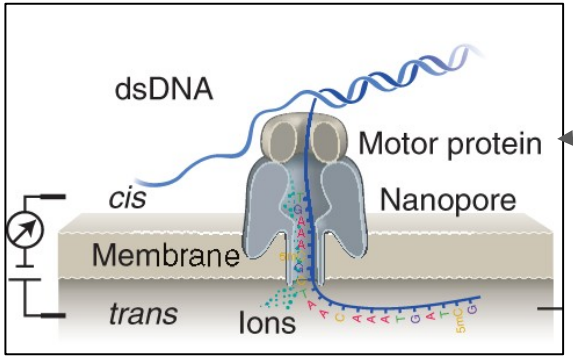


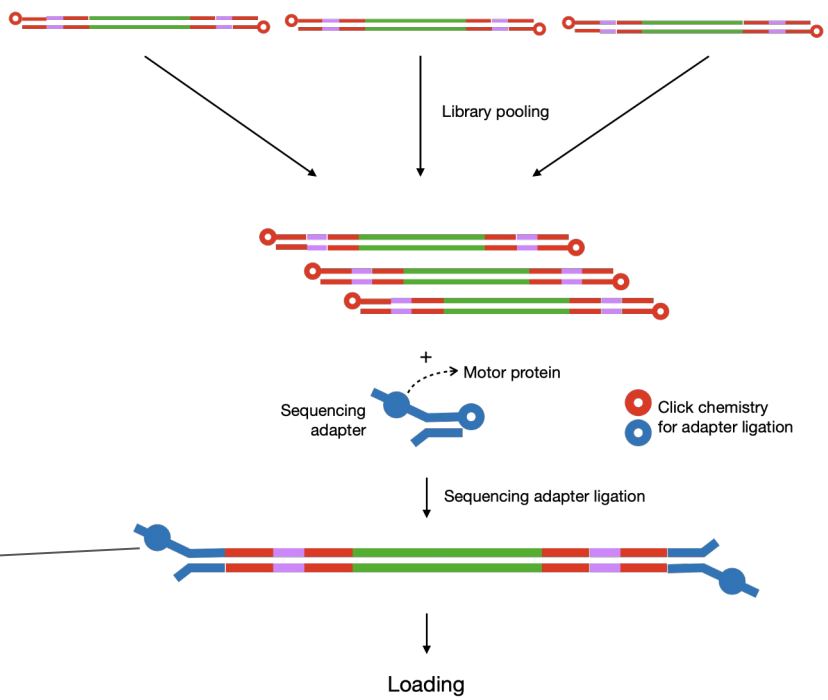
RNA mNGS library prep for ONT (SMART-9N)

STEP 5. Sequencing adapter ligation

(post-PCR lab, immediately before sequencing)

- ONT sequencing adapters have a motor protein attached to it that helps guide the DNA through the nanopores
- Ligation of adapter occurs in 5 min using a proprietary



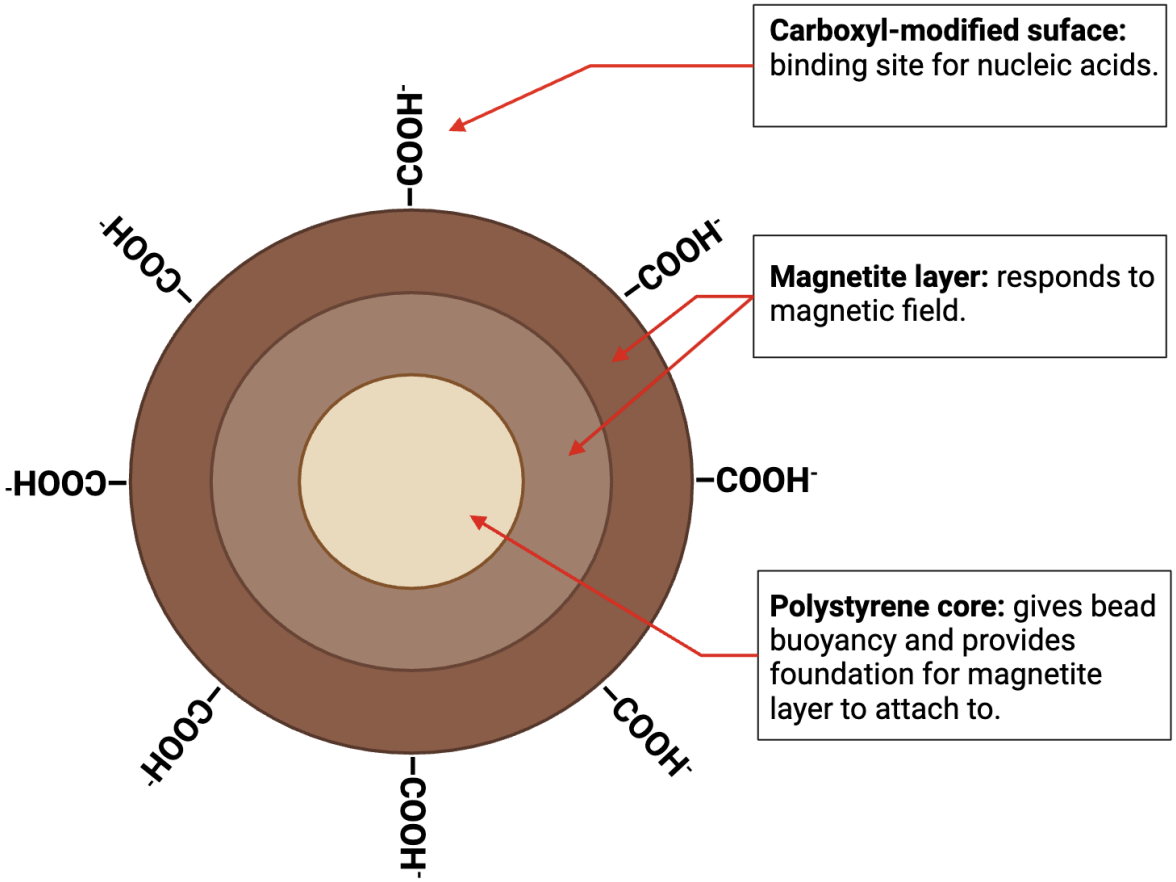


Introduction to SPRI Clean-ups

What are SPRI beads?

- Solid-phase Reversible Immobilization (SPRI) beads are paramagnetic beads used for **purification** and **size selection** of nucleic acids such as:
  - PCR amplicons
  - **NGS libraries**
  - **RNA/DNA**

What are SPRI beads?



How do SPRI beads work? *Purification*

- Under the right conditions, nucleic acids precipitate on the surface of the beads. The following are required:
  - **Crowding agent** to force nucleic acids out of solution.
  - **High salt concentration** to facilitate binding between nucleic acids and beads.
  - **Magnetic field induction** to control movement of nucleic acids.

Contents of SPRI buffer

Content	**Use**
Polyethylene Glycol (PEG) 8000	**Crowding agent** used to minimize space between nucleic acid and beads, favoring their proximity and the nucleic acid precipitating on the surface of the beads.

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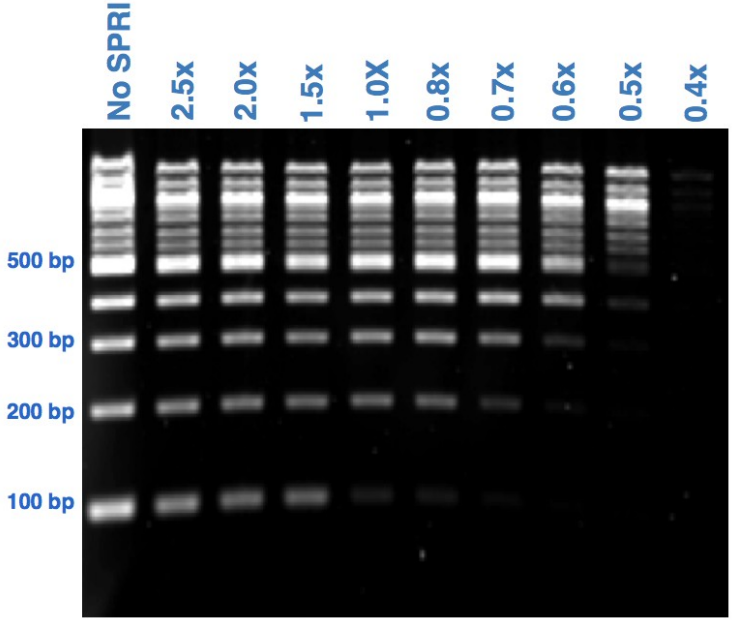
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EDTA	**Chelating agent** which sequesters ions such as Mg2+ and Ca2+ which could fragment/degrade DNA or activate any enzymes (e.g. nucleases) that are present in the sample. Aids in retaining the integrity of the sample.

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EDTA	**Chelating agent** which sequesters ions such as Mg2+ and Ca2+ which could fragment/degrade DNA or activate any enzymes (e.g. nucleases) that are present in the sample. Aids in retaining the integrity of the sample.
Tween 20	**Detergent** which decreases the adhesion of nucleic acids to the plastic walls of the tube by obstructing the ionic attraction.

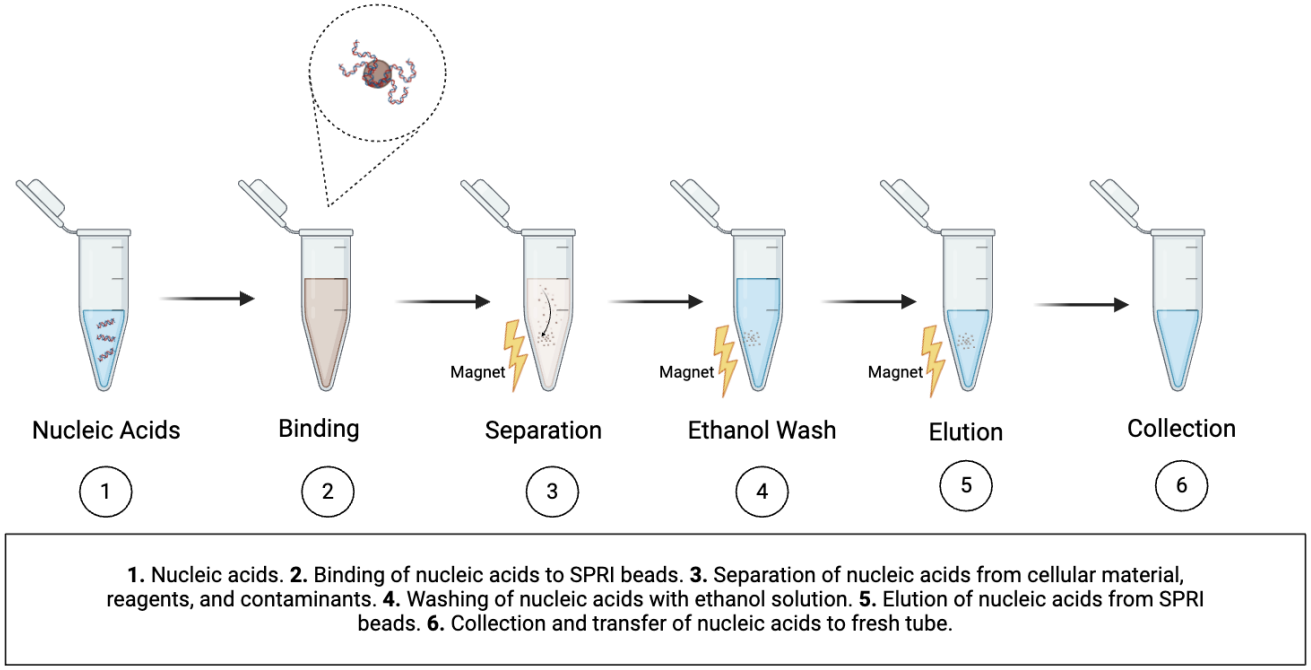
How do SPRI beads work? *Size selection*

- Size selection is dictated by the ratio between the volume of SPRI beads added to your sample and the volume of your sample.
    - **The higher the SPRI bead to sample volume ratio, the more smaller fragments will be retained.**
- For a desired SPRI ratio, to determine the volume of SPRI beads to add to your sample, multiply the SPRI ratio by the volume of your sample.



$$\text{SPRI ratio} = \frac{\text{Volume of beads}}{\text{Volume of reaction}}$$

Using SPRI beads in library preparation: *The process*

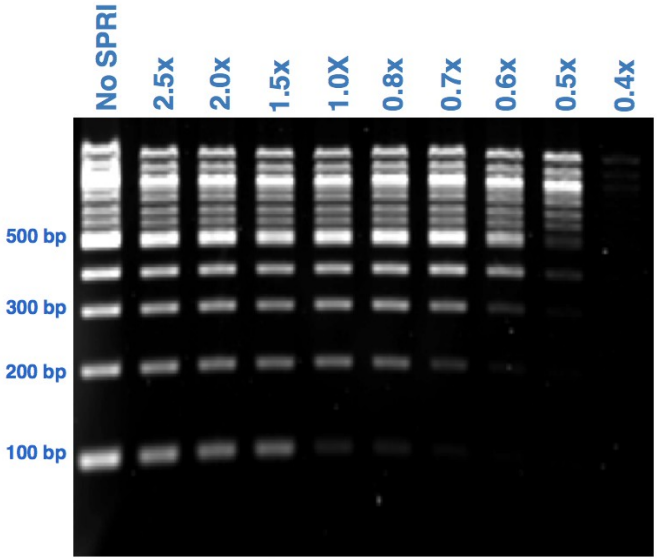


1. Nucleic acids. **2.** Binding of nucleic acids to SPRI beads. **3.** Separation of nucleic acids from cellular material, reagents, and contaminants. **4.** Washing of nucleic acids with ethanol solution. **5.** Elution of nucleic acids from SPRI beads. **6.** Collection and transfer of nucleic acids to fresh tube.

Note: Some commercial options are “SPRISelect Bead-Based Reagent” or “AMPure XP Beads”

Using SPRI beads in library preparation: *At which steps?*

Step	Ideal RNA/library size	To remove	SPRI ratio
After DNAse I treatment	-	Reagents	2x
After barcoding PCR	> 400 bp	Reagents and small fragments / dimers	0.6x
After pooling (if the pool needs to be concentrated	-	To concentrate the pool before loading	2x



$$\text{SPRI ratio} = \frac{\text{Volume of beads}}{\text{Volume of reaction}}$$

SPRI bead cleanup activity

Set up **2** reactions according to the table below at **0.8X** and **2.0X**. DNA input is 50bp DNA ladder diluted to 1:50 (save an aliquot to use as a control).

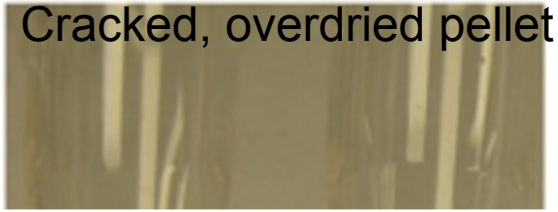
1. Add SPRI beads to sample and mix thoroughly by pipetting up and down ten times. Incubate for 3-5 mins at room temperature, then pulse spin the tubes for 1-2 seconds,
2. Place samples on magnetic rack and incubate for ~5 mins on the rack.
3. Remove supernatant.
4. Add 200µL of 85% EtOH to samples while on the magnetic rack. Do not disturb beads. Incubate at room temperature for 30s then remove the supernatant.
5. Repeat EtOH wash step for a total of 2 washes.
6. Air dry the beads for *up to* 3-5 mins while on the magnetic rack. *Watch beads carefully throughout this incubation to avoid overdrying; smaller bead pellets dry faster. Trust your eyes more than the timer.*
7. Remove tube from magnetic rack. Add 20ul nuclease free water to elute.
8. Pipette to mix well at least ten times. Incubate for a minimum of 2 mins at room temperature off the magnetic rack, then pulse spin the tubes for 1-2 seconds.
9. Place on magnetic rack until solution is clear ~5 mins.
10. Collect 18ul of the supernatant and transfer to a clean nuclease free PCR tube

Bead:Sample	0.8 : 1	2 : 1
50bp ladder 1:50 (µL)	20	20
SPRI beads (µL)	16	40

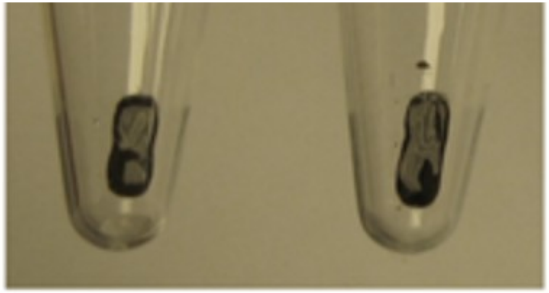
Do **not** overdry SPRI beads

Shiny, wet pellet

Cracked, overdried pellet

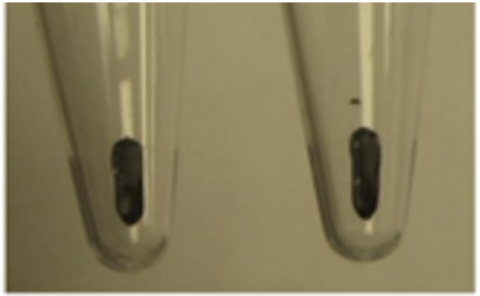


Risk of **EtOH carryover**





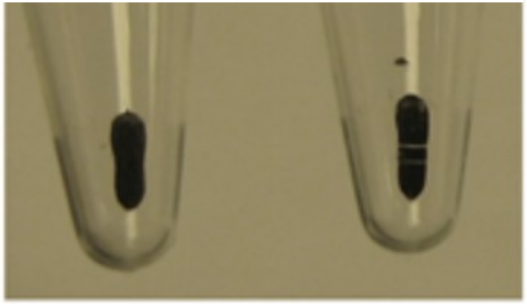
Ready for **elution**



Matte, dry pellet



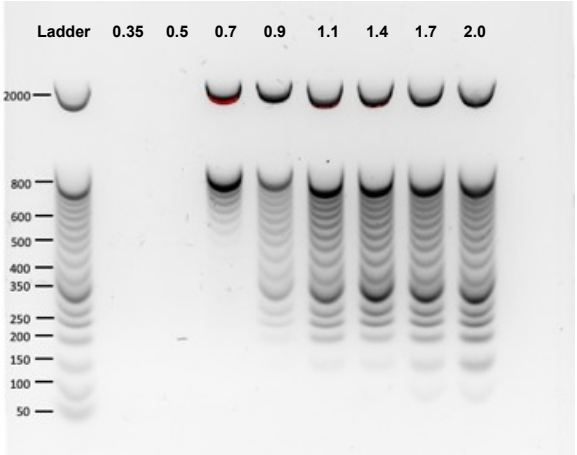
Risk of **low yield**



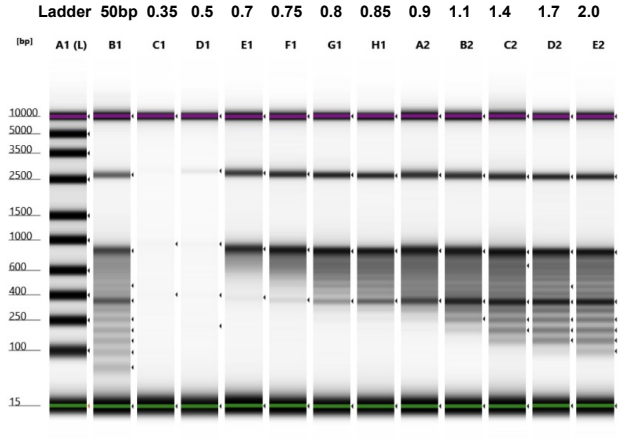
SPRI bead cleanup results

Quantify recovered sample by Qubit dsDNA HS

Visualize samples by agarose gel electrophoresis (1.8% agarose gel and 10000x SYBR GOLD DNA Stain or SYBR-Safe stain), capillary electrophoresis, or automated electrophoresis (TapeStation).



1.8% Agarose



TapeStation

Next : 05-O_Library prep for ONT QC